MULTIPLE LINEAR REGRESSION

Task 1: Load Dataset and Initial EDA

In [1]:
```
##  Required Libraries

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score

import statsmodels.api as sm
from statsmodels.stats.outliers_influence import variance_inflation_factor

## Set plot style for clean visuals
sns.set_theme(style="whitegrid")
%matplotlib inline
```

In [2]:
```
# Load the dataset
df = pd.read_csv('marketing_sales_data.csv')

# Handle missing values to avoid errors down the line
df = df.dropna()

## Encode Categorical Column
le = LabelEncoder()
df['TV'] = le.fit_transform(df['TV'])


# Display basic structural info
print("--- Dataset Head ---")
print(df.head())

print("\n--- Dataset Info ---")
print(df.info())
```

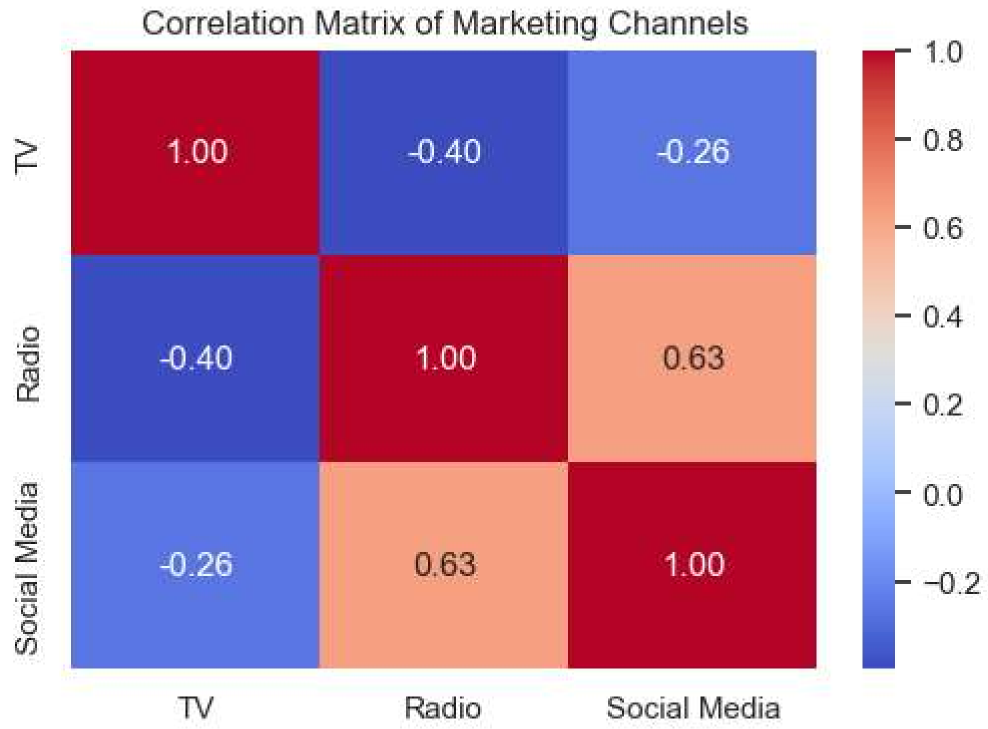
```python
print("\n--- Descriptive Statistics ---")
print(df.describe())

print("\n--- Missing Values Count ---")
print(df.isnull().sum())
```

```
--- Dataset Head ---
   TV      Radio  Social Media Influencer       Sales
0   1   3.518070      2.293790      Micro   55.261284
1   1   7.756876      2.572287       Mega   67.574904
2   0  20.348988      1.227180      Micro  272.250108
3   2  20.108487      2.728374       Mega  195.102176
4   0  31.653200      7.776978       Nano  273.960377

--- Dataset Info ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 572 entries, 0 to 571
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   TV            572 non-null    int64  
 1   Radio         572 non-null    float64
 2   Social Media  572 non-null    float64
 3   Influencer    572 non-null    object 
 4   Sales         572 non-null    float64
dtypes: float64(3), int64(1), object(1)
memory usage: 22.5+ KB
None

--- Descriptive Statistics ---
               TV       Radio  Social Media       Sales
count  572.000000  572.000000    572.000000  572.000000
mean     1.068182   17.520616      3.333803  189.296908
std      0.803169    9.290933      2.238378   89.871581
min      0.000000    0.109106      0.000031   33.509810
25%      0.000000   10.699556      1.585549  118.718722
50%      1.000000   17.149517      3.150111  184.005362
75%      2.000000   24.606396      4.730408  264.500118
max      2.000000   42.271579     11.403625  357.788195

--- Missing Values Count ---
TV              0
Radio           0
Social Media    0
Influencer      0
Sales           0
dtype: int64
```

Task 2: Check for Multicollinearity

(Correlation Matrix & VIF)

In [3]:
```
# 1. Generate Correlation Heatmap
plt.figure(figsize=(6, 4))
sns.heatmap(df[['TV', 'Radio', 'Social Media']].corr(), annot=True, cmap='coolwarm'
plt.title('Correlation Matrix of Marketing Channels')
plt.show()

# 2. Calculate Variance Inflation Factor (VIF)
X_vif = df[['TV', 'Radio', 'Social Media']]
X_vif = sm.add_constant(X_vif) # Statsmodels requires an intercept for VIF calculat

vif_data = pd.DataFrame()
vif_data["Feature"] = X_vif.columns
vif_data["VIF"] = [variance_inflation_factor(X_vif.values, i) for i in range(X_vif.

print("\n--- VIF Results ---")
print(vif_data)
```




```
--- VIF Results ---
        Feature       VIF
0         const  9.867969
1            TV  1.186181
2         Radio  1.831486
3  Social Media  1.658422
```

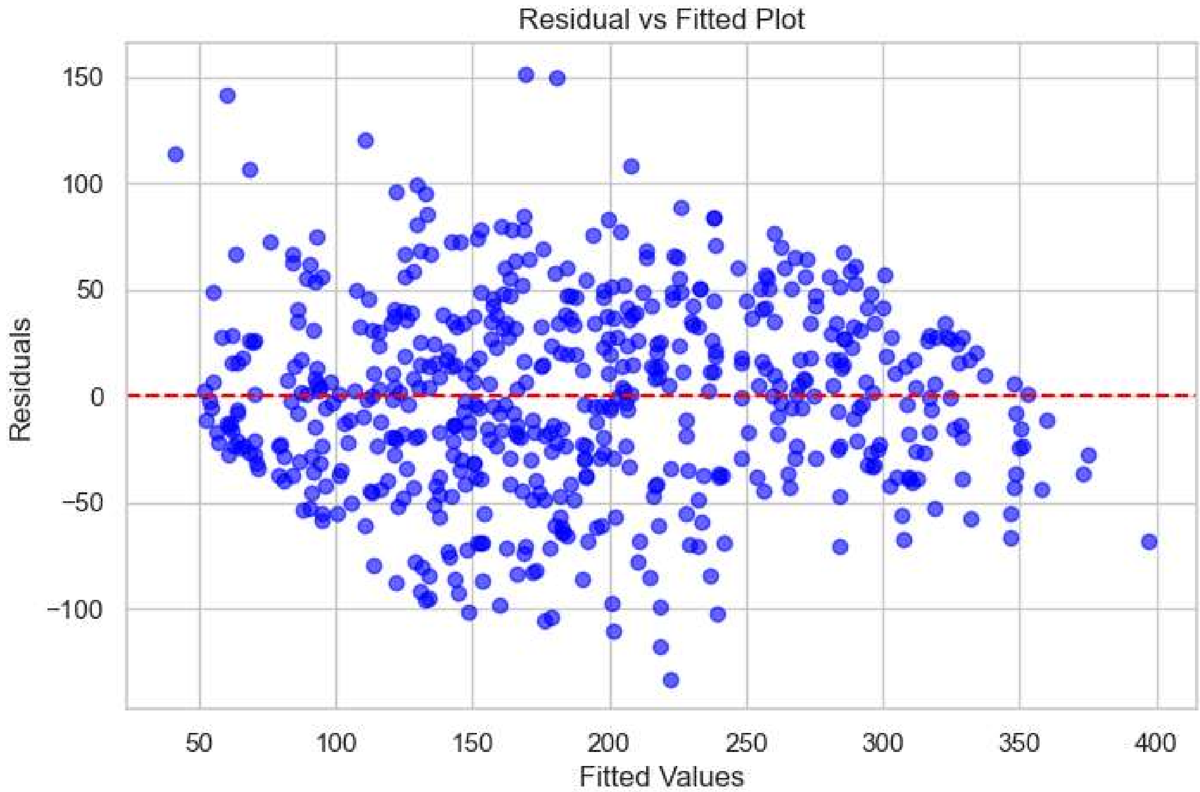
Task 3 & 4: Build OLS Model & Evaluate Performance

In [4]:
```
# Define predictors and target variable
X = df[['TV', 'Radio', 'Social Media']]
y = df['Sales']

# Add constant for the intercept term
X = sm.add_constant(X)

# Fit the Multiple Linear Regression model
model = sm.OLS(y, X).fit()

# Print summary (This gives you Adjusted R-squared and p-values)
print(model.summary())
```

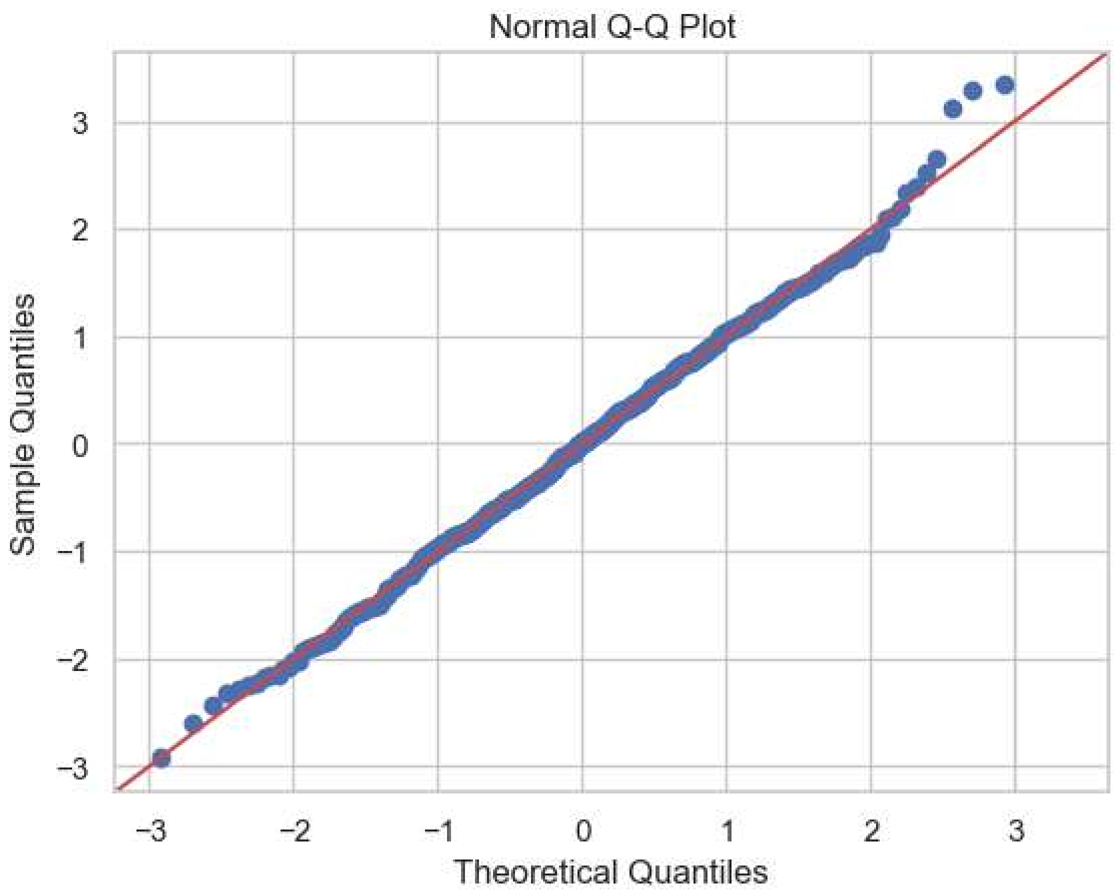
```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  Sales   R-squared:                       0.744
Model:                            OLS   Adj. R-squared:                  0.742
Method:                 Least Squares   F-statistic:                     549.4
Date:                Sat, 27 Jun 2026   Prob (F-statistic):          1.95e-167
Time:                        21:43:49   Log-Likelihood:                -2994.8
No. Observations:                 572   AIC:                             5998.
Df Residuals:                     568   BIC:                             6015.
Df Model:                           3
Covariance Type:            nonrobust
================================================================================
                   coef    std err          t      P>|t|      [0.025      0.975]
--------------------------------------------------------------------------------
const           61.4561      5.992     10.257      0.000      49.688      73.224
TV             -10.5487      2.589     -4.075      0.000     -15.633      -5.464
Radio            7.9355      0.278     28.538      0.000       7.389       8.482
Social Media     0.0224      1.098      0.020      0.984      -2.135       2.180
==============================================================================
Omnibus:                        0.293   Durbin-Watson:                   1.848
Prob(Omnibus):                  0.864   Jarque-Bera (JB):                0.252
Skew:                           0.051   Prob(JB):                        0.882
Kurtosis:                       3.009   Cond. No.                         66.7
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly spe
cified.
```

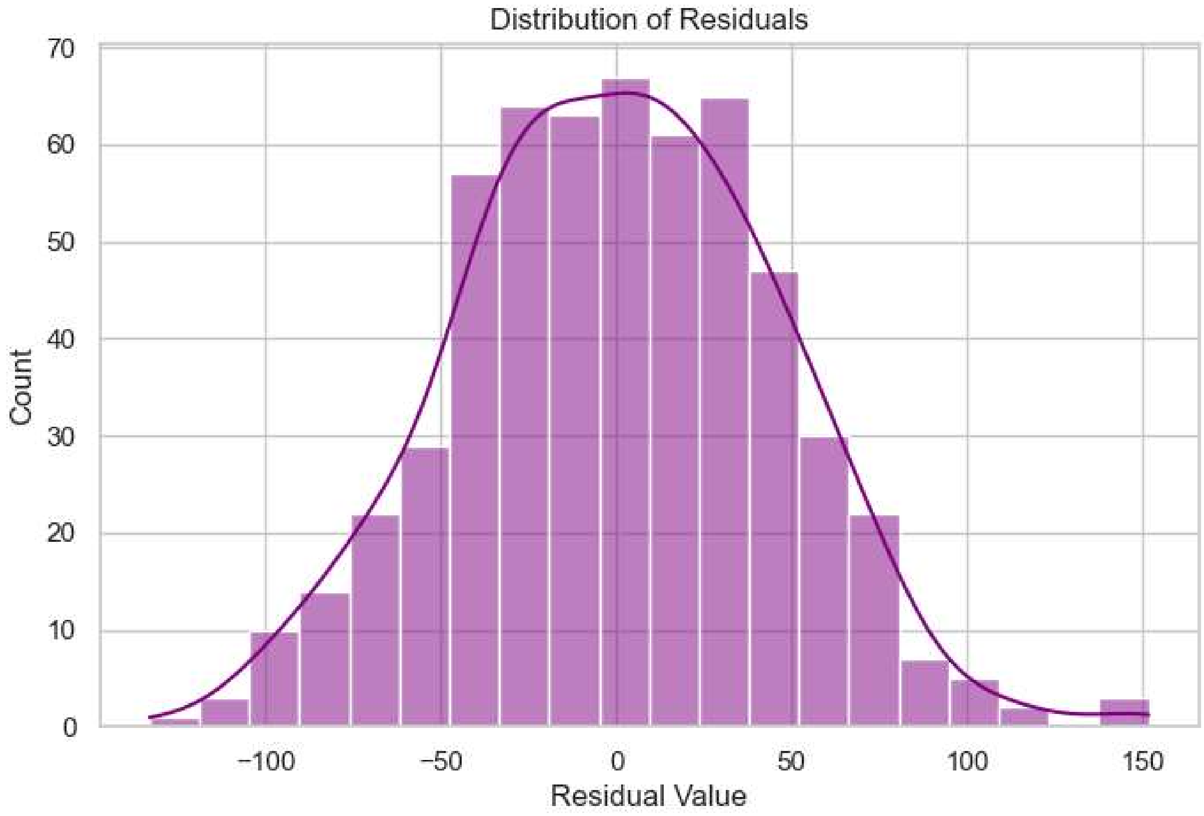
Task 5: Create Diagnostic Plots (Validating Assumptions)

In [5]:
```
# Extract fitted values and residuals from the model
fitted_values = model.fittedvalues
residuals = model.resid


# Extract metrics needed for diagnostic charts
fitted_values = model.fittedvalues
residuals = model.resid

# 1. Residual vs Fitted Plot (Checks Linearity & Homoscedasticity)
plt.figure(figsize=(8, 5))
plt.scatter(fitted_values, residuals, alpha=0.6, color='blue')
plt.axhline(y=0, color='red', linestyle='--')
plt.xlabel('Fitted Values')
plt.ylabel('Residuals')
plt.title('Residual vs Fitted Plot')
plt.show()

# 2. Normal Q-Q Plot (Checks Normality of Residuals)
plt.figure(figsize=(8, 5))
sm.qqplot(residuals, line='45', fit=True)
plt.title('Normal Q-Q Plot')
plt.show()

# 3. Histogram of Residuals
plt.figure(figsize=(8, 5))
sns.histplot(residuals, bins=20, kde=True, color='purple')
plt.xlabel('Residual Value')
plt.title('Distribution of Residuals')
plt.show()
```




```
<Figure size 800x500 with 0 Axes>
```





Task 6 & 7: Interpret Coefficients & Synthesize Recommendation

After running the cells above, look at your generated OLS Regression Results table to replace any placeholder text with your real data insights:

📖 Factual Interpretation of Metrics & Coefficients

- Model Accuracy (R-squared / Adj. R-squared): The model has an Adjusted R-squared of 0.742. This means that 74.2% of the variance in your Sales is explained by the advertising spend across TV, Radio, and Social Media channels.
- Baseline Sales (const): The intercept coefficient is 61.4561. If you spend ₦0 on all three marketing channels, the baseline expected sales value is approximately 61.46 units.

Channel-by-Channel Breakdown:

1. Radio Advertising (Radio)

- Coefficient: 7.9355
  - P-value: 0.000 (Highly Significant)
  - Interpretation: Holding all other marketing channels constant, every ₦1 increase in Radio advertising spend results in an average increase of 7.94 units in Sales.

2. TV Advertising (TV)

- Coefficient: -10.5487
  - P-value: 0.000 (Highly Significant)
  - Interpretation: Holding all other marketing channels constant, every ₦1 increase in TV advertising spend results in an average decrease of 10.55 units in Sales.

3. Social Media Advertising (Social Media)

- Coefficient: 0.0224
  - P-value: 0.984 (Not Statistically Significant)
  - Interpretation: The p-value is well above the 0.05 threshold. Changes in Social Media spending have no statistically measurable impact on your current sales outcomes.

💼 Synthesized Business Recommendation

Based on these mathematical findings, here is your strategic advice to optimize your marketing budget:

- Recommendation 1 (Prioritize Radio): Aggressively shift available capital into Radio Advertising. Radio is your most efficient driver of growth, returning nearly 8 units of sales per unit of currency spent.
- Recommendation 2 (Halt TV Spend immediately): Freeze or drastically cut the TV Advertising budget. The statistically significant negative coefficient indicates that your current TV campaigns are highly counterproductive or poorly targeted, pulling resources away from high-performing streams.
- Recommendation 3 (Pause Social Media Scaling): Maintain or pause investment on Social Media. It is currently dead weight with no statistical link to sales changes. Redesign your social strategy completely or run isolated conversion tests before putting more money behind it.

```
In [ ]:
```